

Figure 10.10 Variation of displacement, velocity and acceleration at different instant of time

(y = 0), velocity of the particle is maximum but the acceleration of the particle is zero. At the extreme position ($y = \pm A$), the velocity of the particle is zero but the acceleration is maximum $\mp A\omega^2$ acting in the opposite direction.

EXAMPLE 10.3

Which of the following represent simple harmonic motion?

(i) $x = A \sin \omega t + B \cos \omega t$

(ii) $x = A \sin \omega t + B \cos 2\omega t$

(iii) $x = A\, e^{i\omega t}$

(iv) $x = A \ln \omega t$

Solution

(i) $x = A \sin \omega t + B \cos \omega t$

$$\frac{dx}{dt} = A\,\omega \cos \omega t - B\,\omega \sin \omega t$$

$$\frac{d^2x}{dt^2} = -\,\omega^2\,(A \sin \omega t + B \cos \omega t)$$

$$\frac{d^2x}{dt^2} = -\,\omega^2 x$$

This differential equation is similar to the differential equation of SHM (equation 10.10).

Therefore, $x = A \sin \omega t + B \cos \omega t$ represents SHM.

(ii) $x = A \sin \omega t + B \cos 2\omega t$

$$\frac{dx}{dt} = A\,\omega \cos \omega t - B\,(2\omega) \sin 2\omega t$$

$$\frac{d^2x}{dt^2} = -\,\omega^2\,(A \sin \omega t + 4B \cos 2\omega t)$$

$$\frac{d^2x}{dt^2} \neq -\,\omega^2 x$$

This differential equation is not like the differential equation of a SHM (equation 10.10). Therefore, $x = A \sin \omega t + B \cos 2\omega t$ does not represent SHM.

(iii) $x = A\, e^{i\omega t}$

$$\frac{dx}{dt} = A\, i\omega e^{i\omega t}$$

$$\frac{d^2x}{dt^2} = -A\,\omega^2\, e^{i\omega t} = -\omega^2 x \qquad (\because i^2 = -1)$$

This differential equation is like the differential equation of SHM (equation 10.10). Therefore, $x = A\, e^{i\omega t}$ represents SHM.

(iv) $x = A \ln \omega t$

$$\frac{dx}{dt} = \left(\frac{A}{\omega t}\right)\omega = \frac{A}{t}$$

$$\frac{d^2x}{dt^2} = -\frac{A}{t^2} \Rightarrow \frac{d^2x}{dt^2} \neq -\omega^2 x$$

This differential equation is not like the differential equation of a SHM (equation 10.10). Therefore, $x = A \ln \omega t$ does not represent SHM.

EXAMPLE 10.4

Consider a particle undergoing simple harmonic motion. The velocity of the particle at position x_1 is v_1 and velocity of the particle at position x_2 is v_2. Show that the ratio of time period and amplitude is

$$\frac{T}{A} = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 x_2^2 - v_2^2 x_1^2}}$$

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